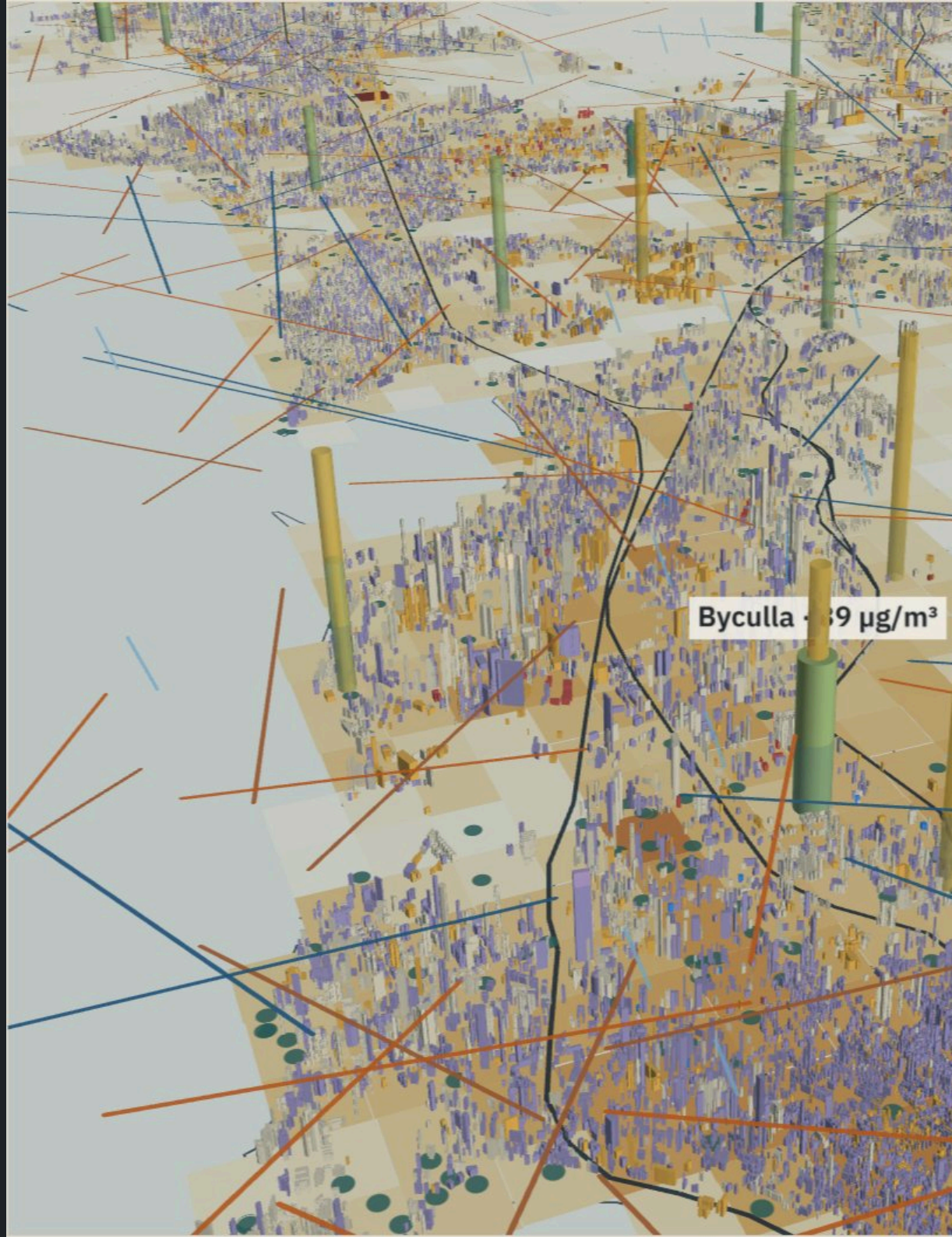
PROJECT GUIDE

Mumbai: The Season of Smog

Why the city's air turns bad after the monsoon, shown on a 3D model built from real BMC buildings.

GOOGLE MUMBAI DATA CHALLENGE 2026

Most Creative Submission



THE QUESTION

Why does Mumbai's air turn bad every winter?

The city doesn't change in October. The air does. Between the monsoon and October–November, four things shift at once.

WIND FROM THE LAND

5% → 59%

of hours. Inland air replaces sea air.

RAIN PER MONTH

528 → 45 mm

Nothing washes particles out.

NIGHT MIXING LAYER

442 → 180 m

Same smoke, less air to mix into.

PM2.5, MEDIAN

14 → 48 µg/m³

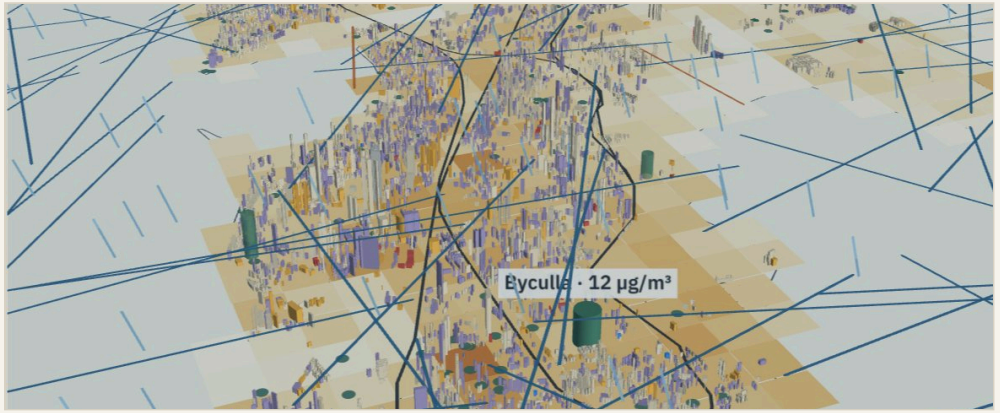
3.5× dirtier. Measured at CPCB monitors.

The short answer

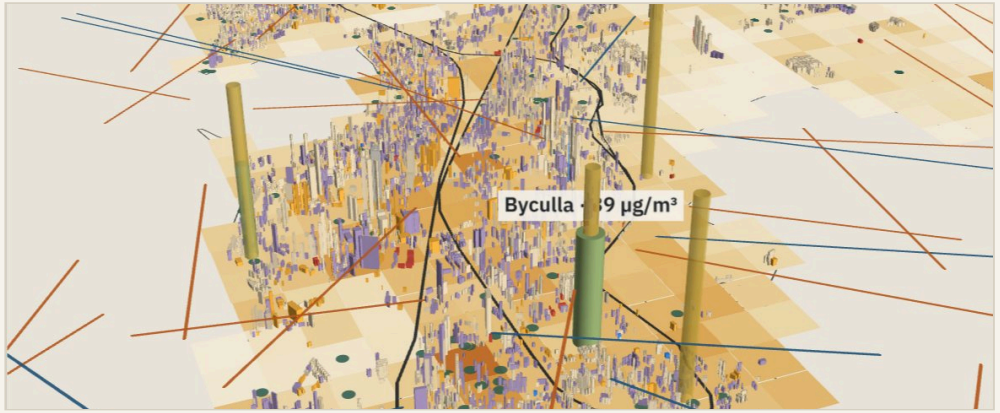
Emissions from roads, building sites, industry and burning carry on all year. In the monsoon, wind off the sea and heavy rain clear them. After it, the wind turns, the rain stops and a shallow layer of air near the ground holds everything in.

Is it getting worse?

Not established. Only two winters are near-complete in the record, measured at different stations. The season is severe and recurring, but this data can't show a trend over the years.



Monsoon (Jun–Sep). The same buildings, wind off the sea, heavy rain.



Now (Oct–Nov). Wind off the land, little rain, a low lid.

HOW THE PAGE IS ORGANISED

Everything is on one scrolling page

1. **The headline and ENTER THE CITY.** The button opens the full-screen walkthrough.
2. **The 3D city.** Pick a season and a reason, or fly into a monitor's neighbourhood.
3. **Why now.** A before-and-after strip and the Plan your day chart.
4. **Full evidence.** Folded away. Calendar, weather, trend and model charts.
5. **Method, sources and limits.** Folded away at the bottom.

Three kinds of number

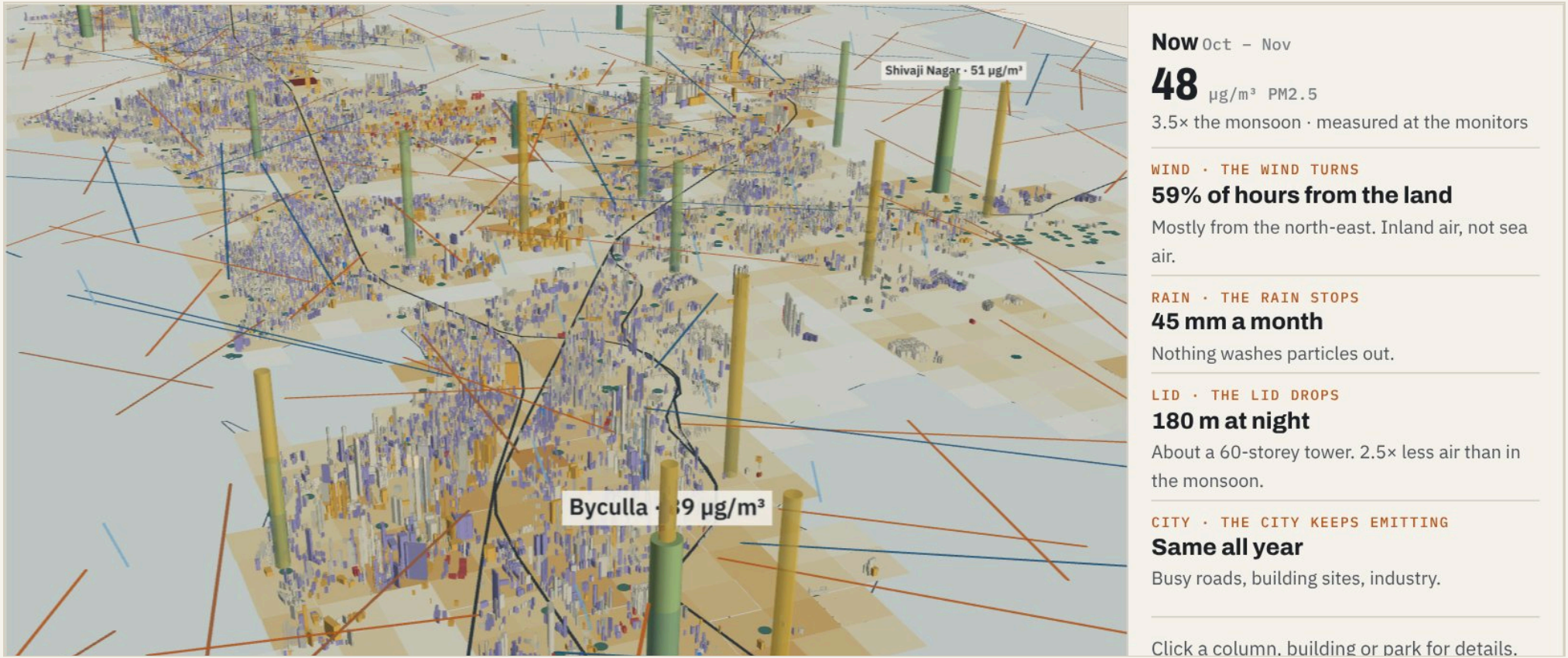
Measured: CPCB PM2.5 readings.
Gridded weather: ERA5 wind, rain and mixing height, one cell over Mumbai.
Derived or research: indicators computed from BMC data, and published emission inventories. Every chart is tagged.



The opening screen. The pointing hand nudges toward the button.

THE 3D CITY

Pick a season and a reason, and watch it on the city



Whole-city view, October–November, wind reason selected. Columns are measured PM2.5 at each monitor; streams show the wind that season.

View

Whole city, or fly into one of the four monitor neighbourhoods.

Season · Reason

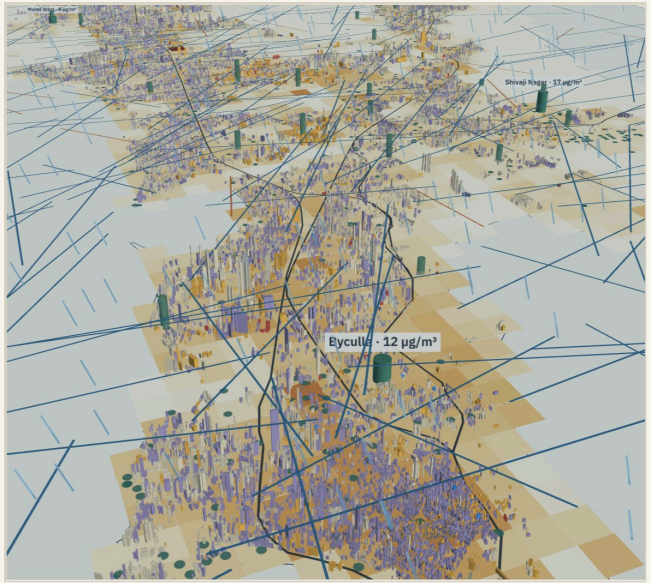
Monsoon, Now, Winter, Summer. Wind, Rain, The lid, or What the city emits.

Click

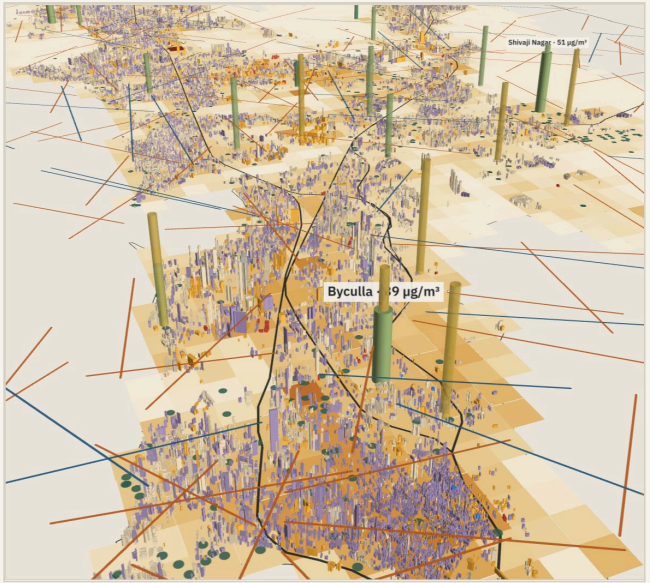
Any column, building, park or stop for its record. Drag to rotate, scroll to zoom.

THE REASONS

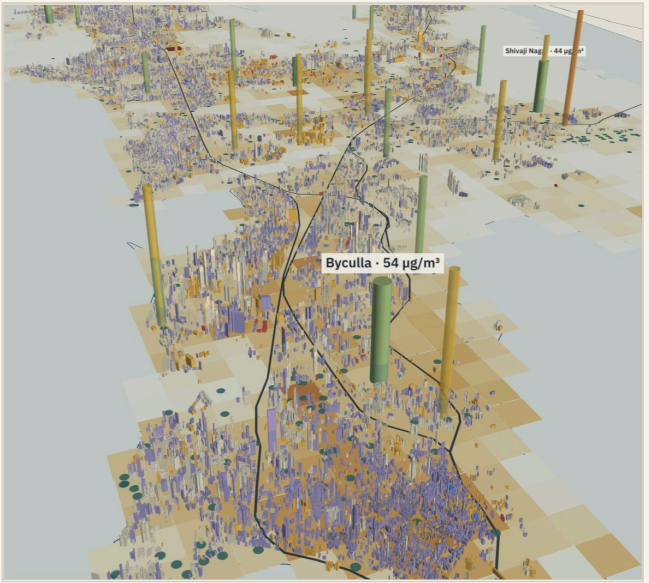
The same city in three seasons



Monsoon. Wind from the sea, heavy rain, clean air.



Now. Wind from the land, little rain, PM2.5 up.

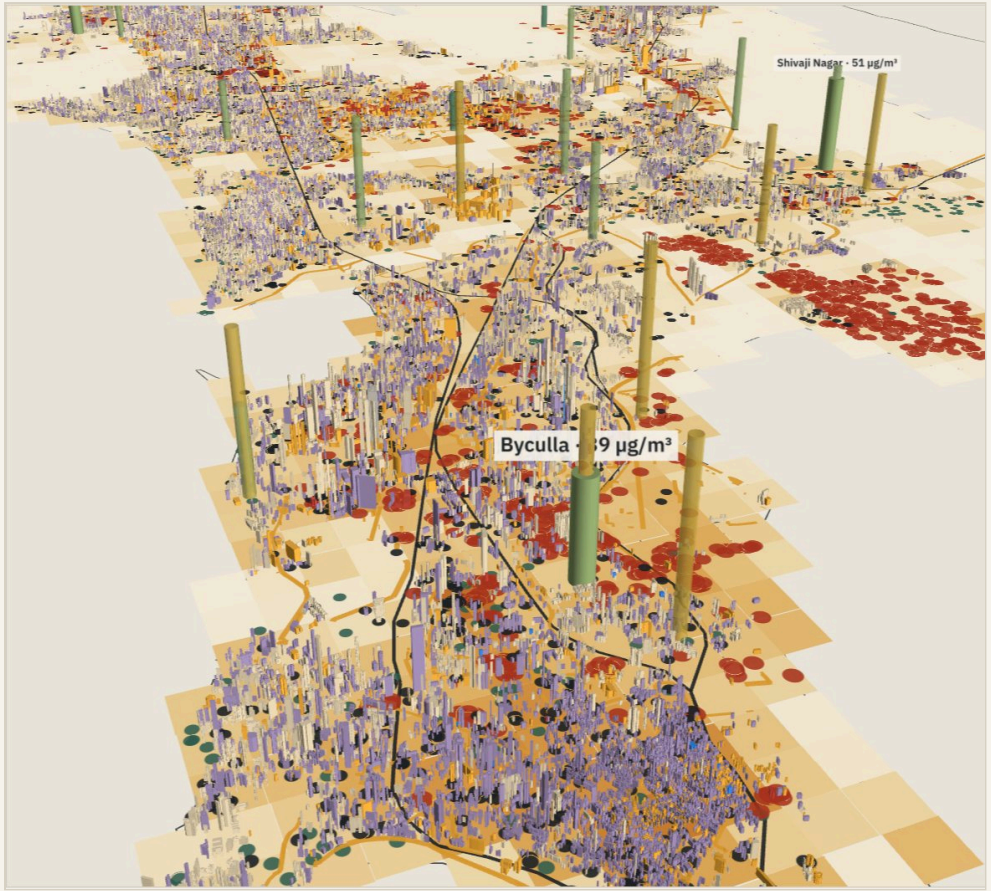


Winter. The night-time mixing layer drops to 109 m (the blue sheets).

	MONSOON Jun – Sep	NOW Oct – Nov	WINTER Dec – Feb	SUMMER Mar – May
Wind from the land (% of hours)	5	59	61	28
Rain (mm a month)	528	45	0	34
Night mixing layer (m)	442	180	109	263
PM2.5, measured median (µg/m³)	14	48	50	23

Wind streams, rain and the blue mixing-height sheets are drawn from ERA5 weather data for each season. They show the weather, not pollution. PM2.5 appears only as the measured columns at the monitors.

What stays the same all year, and what is on the ground



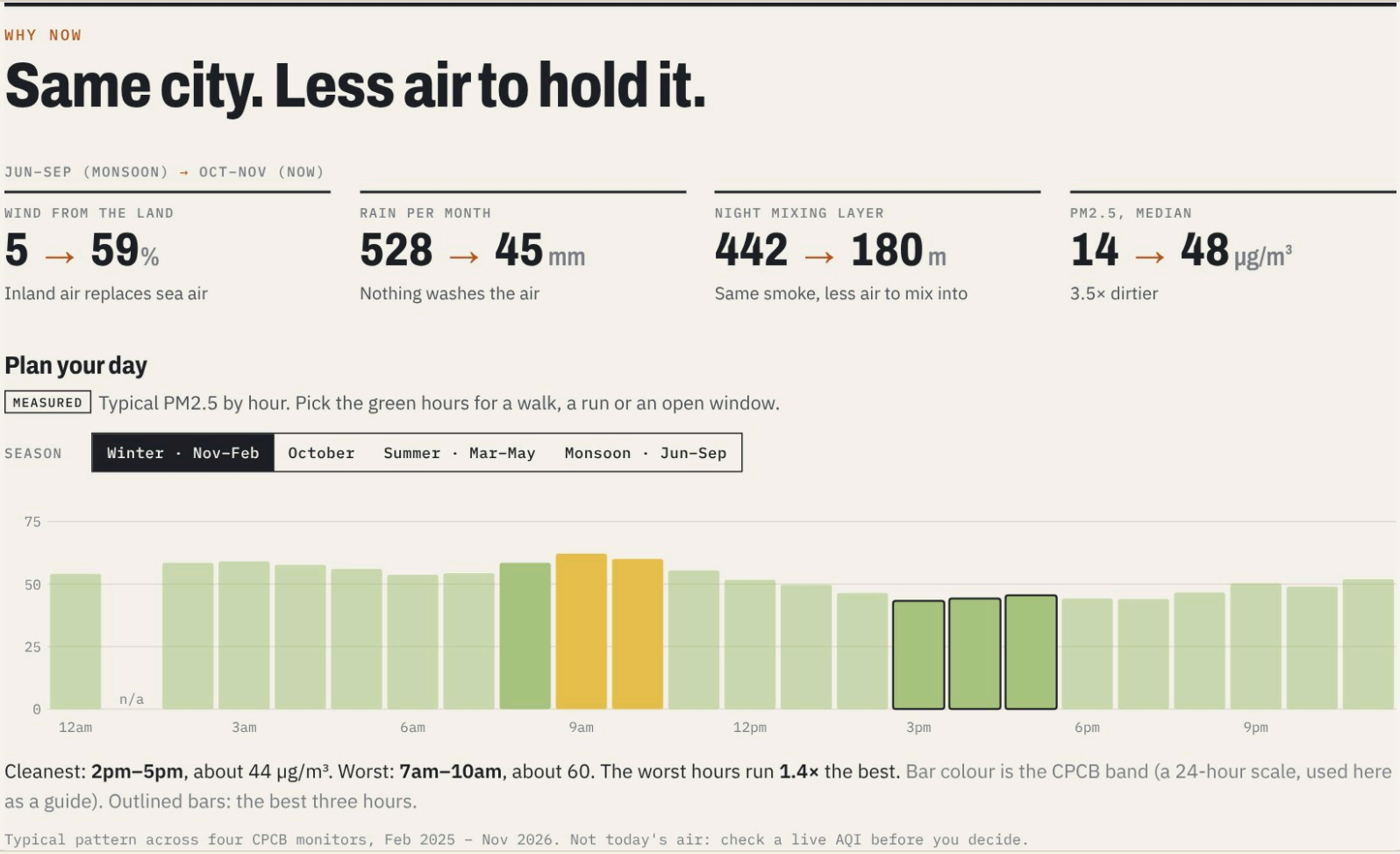
What the city emits. Amber: streets 12 m or wider. Dark dots: 4,421 buildings registered as under construction. Red dots: 2,053 industrial premises. None of this changes with the season.



A monitor's neighbourhood. About 5,000 real BMC buildings within 1.2 km of a monitor, with a month slider that recolours the monitor with its measured PM2.5.

WHY NOW

Same city, less air to hold it, and a chart to plan around it



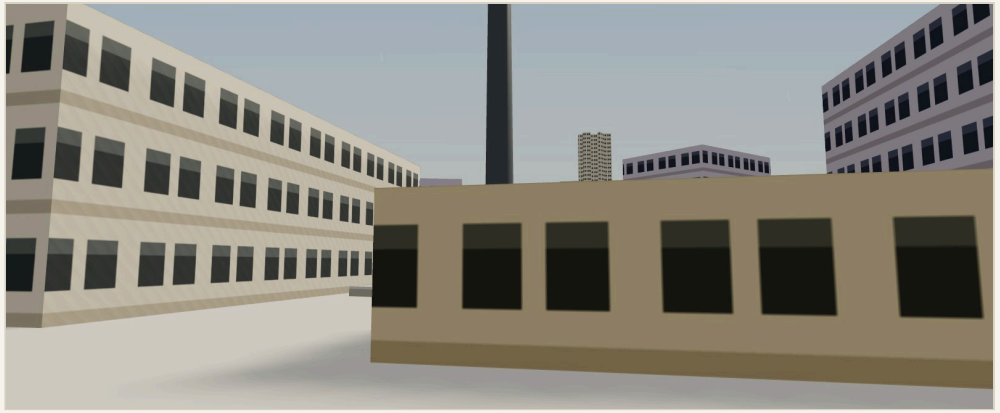
The before-and-after strip, and Plan your day: typical PM2.5 for each hour, by season, with the three best hours outlined.

ENTER THE CITY

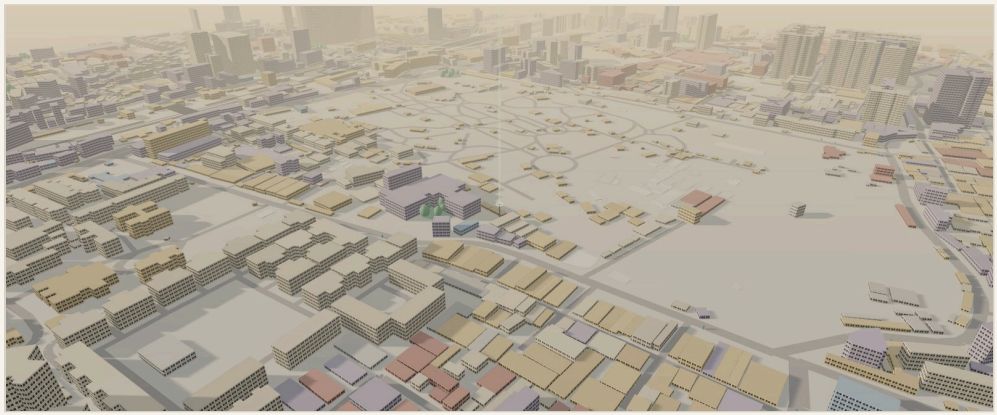
A walk through a real neighbourhood



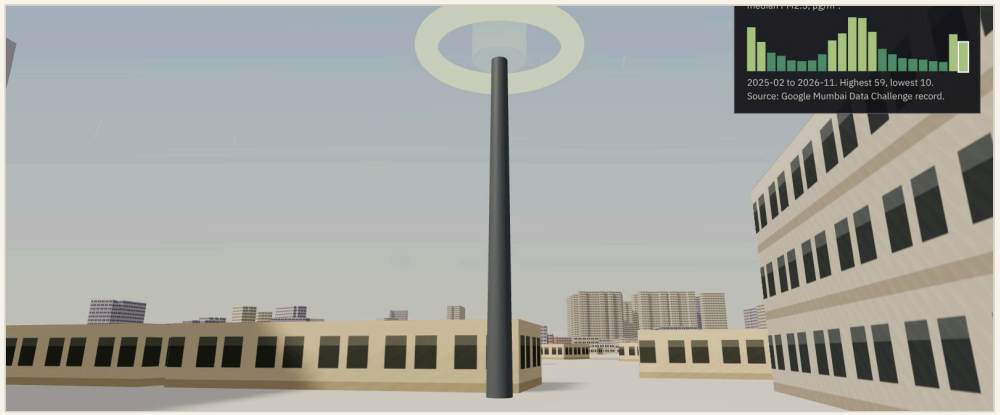
Opening sequence (about 34 s). The camera descends from above the neighbourhood to the monitor.



Walk at eye height. Buildings block you.



Orbit in winter: haze, low sun, PM2.5 for the selected month.



Click the monitor for its full monthly record.

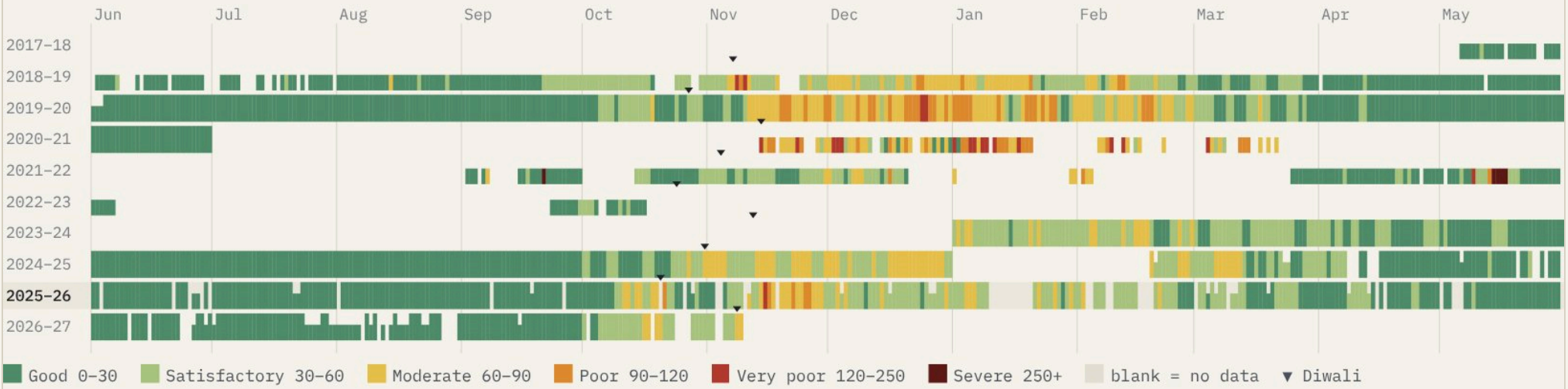
Walk	W A S D · drag to look · Shift to run	**Fly**	W A S D · E / Q up and down	**Orbit**	drag · scroll to zoom
Season	Monsoon · Post-monsoon · Winter	**Month**	slider over the real record	**Inspect**	click buildings, roads, parks, stops, stations

FULL EVIDENCE · THE RECORD

Eight monsoons of data

The pollution calendar

MEASURED One row per year, June to May. Each cell is a day. Click a row to open it below.



PM2.5 µg/m³, daily, coloured by CPCB AQI band. Archives: CPCB-derived archive (Kaggle), 2018–20 (10 stations, 748 days); OpenAQ legacy samples, 2020–22 (2 stations, 267 days); Challenge record, 2025–26 (4 stations, 558 days); XKDR archive, 2024 (29 stations, 366 days). Archives use different stations and are not directly comparable.

- **Every day since 2018** coloured by CPCB band. Blank means no data, never an estimate.
- **The record is broken.** Four archives with different stations, and nothing at all between late 2022 and the end of 2023.
- **Diwali** day ran 1.3 to 2.9 times the weeks around it: a short spike, not the reason the season is bad.

Not established

Winter by winter

MEASURED Greyed rows are incomplete: don't compare them.

WINTER	ARCHIVE	DAYS COVERED	STATIONS	MEDIAN	90TH PCT	95TH PCT	PEAK DAY	HIGH DAYS	LONGEST RUN	ONSET
2018–19	CPCB-derived	96%	1	58	85	94	131 08 Nov	55 48%	16 d	28 Sep
2019–20	CPCB-derived	100%	9	76	101	104	147 25 Dec	90 75%	30 d	12 Oct
2020–21	OpenAQ legacy	57%	2	85	136	161	202 03 Dec	57 84%	10 d	–
2021–22	OpenAQ legacy	47%	2	39	68	77	96 16 Dec	12 21%	3 d	–
2023–24	XKDR	49%	25	53	64	68	87 16 Feb	13 22%	4 d	–
2024–25	Challenge + XKDR	59%	25	62	77	81	88 02 Nov	39 55%	14 d	24 Oct
2025–26	Challenge	81%	4	52	78	88	123 15 Nov	28 29%	11 d	10 Oct

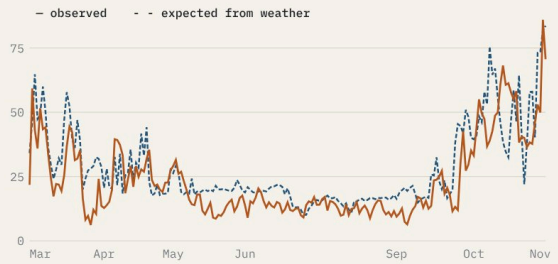
High day = daily PM2.5 above 60 µg/m³ (India's 24-hour standard). Longest run counts consecutive high days with data. Onset = first of five consecutive days at or above 40 µg/m³ after 1 September, where autumn coverage is at least 80%.

- **Only two winters are near-complete:** 2019–20 (median 76 µg/m³, 9 stations) and 2025–26 (52, 4 stations). They aren't comparable.
- Where the same stations appear in both archives, 2024 reads lower than 2019–20 in **24 of 24** station-months.
- The season is severe and recurring. Answering the trend question needs one unbroken series from fixed stations.

What explains this year, and what doesn't

Taking the weather out

STATISTICAL ESTIMATE A model of what weather alone usually brings, tested on data it never saw.



Test: 2026-03-01 to 2026-11-09. Error 7.8 µg/m³ against 11.8 for a naive guess.

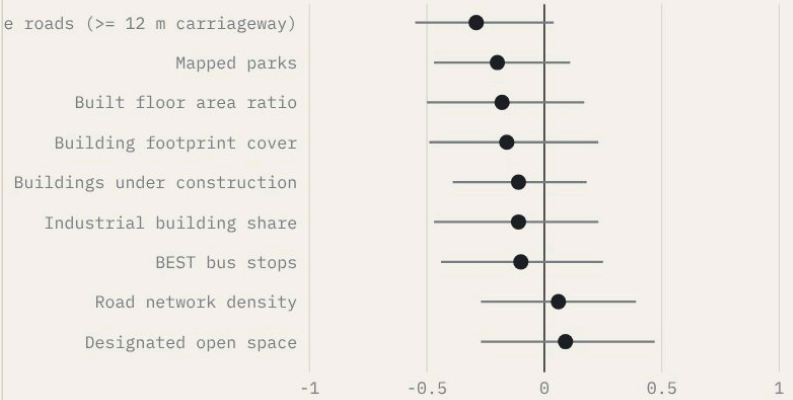
PERIOD	OBSERVED	EXPECTED	RATIO	STATUS
Oct 15 – Nov 10, 2025	43	35	**1.22**	in the training period
Oct 15 – Nov 10, 2026	53	52	**1.01**	held out from training
Winter 2025–26 (Nov–Feb)	54	47	**1.14**	in the training period
Mar – Sep 2026	19	23	**0.80**	held out from training
Nov – Dec 2024	64	50	**1.29**	different archive and stations
Winter 2019–20 (Nov–Feb)	74	39	**1.88**	different archive and stations

Ratio = observed ÷ weather-expected. Above 1: more PM2.5 than this weather usually brought in the training period. It is not a measurement of emissions; it also absorbs anything the model lacks, such as fires, festivals, regional transport and instrument differences.

This autumn: observed **53** against weather-expected **52** µg/m³ (ratio 1.01). About what its weather would predict. Exploratory: one year of training.

Does urban form predict the reading?

STATISTICAL ESTIMATE Correlation with 2024 median PM2.5 (1 km), 25 stations.

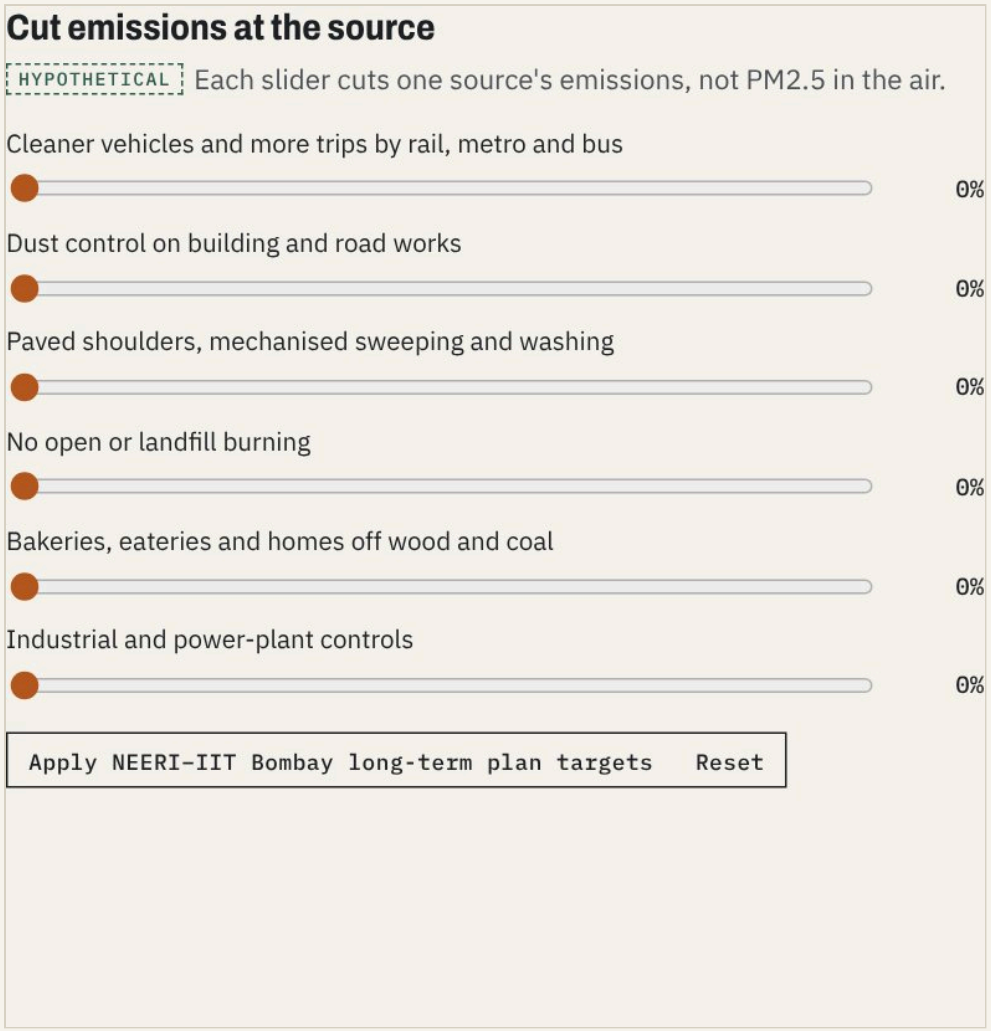


No clear link. Strongest |ρ| is 0.29 and every interval crosses zero. A monitor mostly reads the city's shared air.

- **Weather model.** Trained on Feb 2025 – Feb 2026, tested on Mar – Nov 2026. This autumn: observed **53** against weather-expected **52** µg/m³. Error 7.8 against 11.8 for a naive guess. Exploratory: one year of training.
- **Season does much of the work.** Across the year, daily PM2.5 and the share of hours with land wind correlate at 0.75; inside winter only 0.12.
- **Urban form.** Across 25 stations, no surroundings measure (building cover, roads, bus stops, parks) has a rank correlation stronger than 0.29, and every interval crosses zero.
- **Not claimed:** El Niño or ground-level ozone. Two seasons are too few for the first and the dataset has no ozone readings.

WHAT COULD CHANGE

Cut emissions at the source, and compare two inventories



- Each slider cuts one source: vehicle exhaust, construction, road dust, waste burning, fuel use, industry.
- Results are shown against **two published inventories** (NEERI–IIT Bombay 2020 for all particle mass; SAFAR 2019–20 for PM2.5). They rank sources differently: road dust leads one, vehicle exhaust the other.
- **Emissions are not concentrations.** A 30% cut in what the city emits is not a 30% cut in what a monitor reads. Turning one into the other needs a dispersion model this data can't drive, so the page doesn't try.

DATA, METHOD AND LIMITS

Where every number comes from

SOURCE	USED FOR
CPCB PM2.5, challenge Train.csv and AdditionalHistory archives	All measured values
ERA5 via Open-Meteo (ClimateVerse)	Wind, rain, humidity, mixing height
BMC building footprints	3D city: storeys, use, age
BMC streets, rail, wards, water	Roads, lines, boundaries
OpenCity parks, BEST stops, suburban rail	Access points and markers
NEERI–IIT Bombay 2020; SAFAR 2019–20	Emission shares (research estimates)

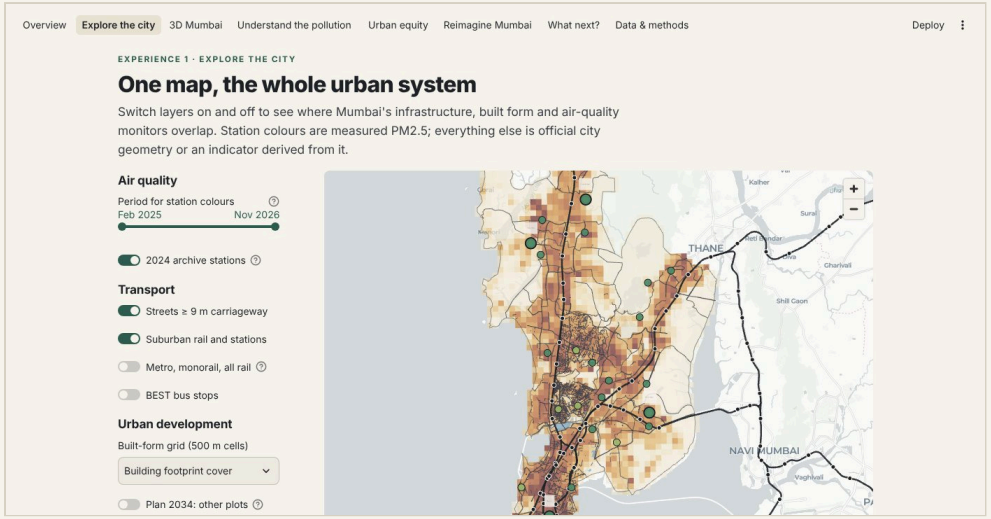
What the page cannot tell you

- Four monitors don't represent every neighbourhood.
- Weather is gridded: one cell over Mumbai, not at the stations.
- Whether winters are worsening over the years.
- Buildings are drawn one by one only within 1.2 km of the monitors; the city view shows buildings of five or more storeys, stretched ×3.
- Scenarios change emissions, not PM2.5.

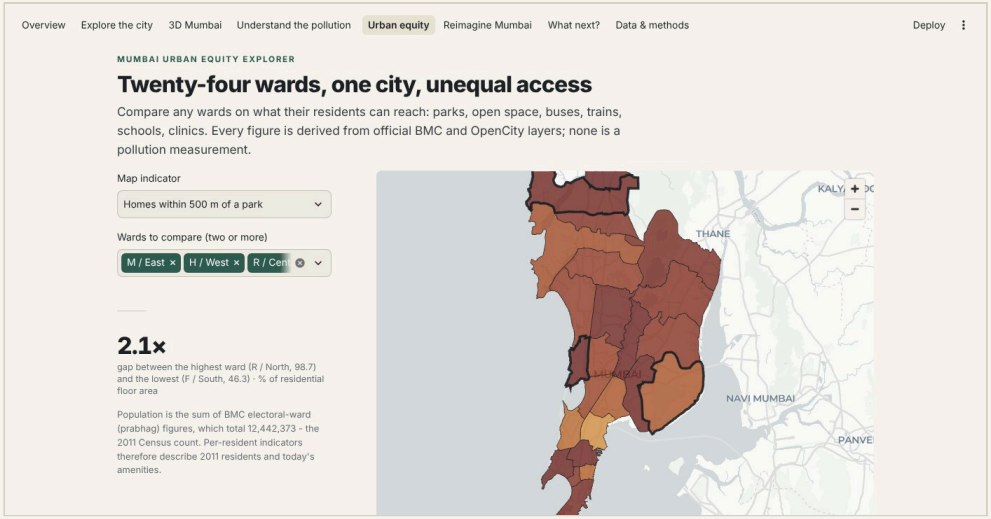
PM2.5 is used as supplied; only the organisers' flagged readings above 1,000 µg/m³ are excluded. Nothing is interpolated.

THE SECOND APP

An eight-page research explorer built in Streamlit



Explore the city. Layered 2D map, station files, and 29 archive stations.



Urban equity. Compare any wards on parks, transit, open space and amenities.

Also in the explorer: 3D buildings around each monitor, pollution storytelling, a scenario lab (build the planned parks, extend bus coverage, change development intensity), evidence-linked recommendations, and a data and methods page.

FILES AND HOW TO RUN

What is in the project

```
mumbai-air-intelligence/
  smog/              the one-page investigation
    index.html       single file, opens anywhere
    build_*.py       data and page builders
    app.js  seasons.js  enter.js
  app.py             eight-page Streamlit explorer
  views/ modules/ visualizations/
  scripts/           fetch_bmc, build_processed, validate
  data/processed/    app-ready tables
  docs/              data audit, demo script, this guide
```

Run it

The one-pager: open `smog/index.html` in a browser (internet needed for the 3D libraries).

The explorer:

```
python -m pip install -r requirements.txt
python -m streamlit run app.py
```

Rebuild the one-pager:

```
./smog/fetch_weather.sh
python smog/build_3d.py
python smog/build_data.py
python smog/build_seasons.py
python smog/build_page.py
```

Raw data goes in `data/raw/` (see its README). The published link is private until its owner shares it.